



Hydrothermal humification and fulvification of grass for artificial humic substance production and by-product applications

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ABSTRACT

This study compares hydrothermal carbonization (HTC), humification (HTH), and fulvification (HTF) of grass biomass to show how increasing alkalinity governs carbon conversion pathways and liquid product formation. Under HTC without alkali addition, carbon conversion was dominated by dehydration and condensation reactions, yielding the highest solid yield (54.10 wt. %) with high carbon content (60.6 wt. %), elevated energy potential (HHV = 25.70 MJ/kg), and the greatest intrinsic thermal stability ($T_{50} = 435\text{ }^{\circ}\text{C}$), while artificial humic acids were not formed and carbon transfer to the liquid phase remained limited (TC = 15.85 g/L; TOC = 15.75 g/L), accompanied by minimal carbon leachability. Moderate alkalinity under HTH reduced solid yield to 44.50 wt% and promoted partial depolymerization and humification, increasing liquid-phase carbon concentrations (TC = 40.47 g/L; TOC = 36.96 g/L) and enabling artificial humic acid formation (2.25 wt. %), while producing hydrochar with favorable surface chemistry for adsorption, reflected by the highest crystal violet uptake (196.28 mg/g). Further increasing alkalinity under HTF shifted carbon partitioning toward the liquid phase, reducing solid yield to 13.18 wt. % while maximizing artificial humic acid formation (8.80 wt. %) and liquid-phase carbon concentrations (TC = 62.15 g/L; TOC = 58.52 g/L); HTF liquid was enriched in lactic acid, characterized by sub-100 nm colloids, the most negative zeta potential (-8.72 mV), and enhanced seed germination up to 95% at 50-fold dilution. Overall, alkalinity is identified as a key parameter directing integrated carbon valorization from energy-dense solids to adsorption-active humified materials and bioactive liquid products within a circular biomass framework.

Abbreviation: A-HSs, Artificial Humic Substances; A-FA, Artificial Fulvic Acid; A-HA, Artificial Humic Acid; ADF, Acid Detergent Fiber; ADL, Acid Detergent Lignin; BET, Brunauer-Emmett-Teller; CV, Crystal Violet; EEM, Excitation Emission Matrix; FTIR, Fourier Transform Infrared Spectroscopy; HHV, Higher Heating Value; HMF, Hydroxymethylfurfural; HS, Humic Substances; HTC, Hydrothermal Carbonization; HTF, Hydrothermal Fulvification; HTH, Hydrothermal Humification; NDF, Neutral Detergent Fiber; OTS, Organic Total Solids; SEM, Scanning Electron Microscopy; TGA, Thermogravimetric Analysis; TC, Total Carbon; TOC, Total Organic Carbon; TS, Total Solids.

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1. Introduction

Grasslands cover a large portion of Europe's agricultural landscape and represent a highly abundant source of biomass, but remain underutilized due to the low economic value of their end-products (Heinrich et al., 2023). Unlike woody biomass or agricultural residues that are already embedded in established bioenergy and material value chains, grass biomass is often produced as a by-product of land management rather than as a primary resource, particularly in permanent grasslands, roadside maintenance, and extensively managed or mature stands. While grasses are mainly harvested for livestock feed, a considerable fraction exhibits low nutritional quality and limited market value, resulting in large quantities of unused or poorly valorized biomass (Meyer et al., 2018). This structurally underexploited biomass stream offers a strategic opportunity for the bioeconomy, as it is continuously available, geographically widespread, does not compete with food production, and requires minimal preprocessing prior to conversion, aligning well with current political and environmental strategies (Spies et al., 2022). Previous studies have found that grasses, particularly monoculture species like *Pennisetum purpureum* (Reza et al., 2020), *Miscanthus giganteus* (elephant grass) (Adesemuyi et al., 2020), and *Poa pratensis* (Kentucky bluegrass) (Trippe et al., 2015), show strong potential for biochar production.

Thermochemical technologies such as pyrolysis and hydrothermal carbonization (HTC) are widely studied for converting lignocellulosic biomass into biochar (Heinrich et al., 2023). Compared to pyrolysis, HTC operates at moderate temperatures (180–260 °C) and allows direct processing of wet feedstocks (Zhang et al., 2019). The resulting hydrochar has been explored for various applications, including environmental remediation, soil improvement (Marzban et al., 2025), and advanced material uses, although its utilization as a solid fuel has historically dominated due to the initial focus on mimicking natural coalification processes (Libra et al., 2011). However, both hydrochar and the associated process liquid may contain phenolic and furanic compounds that limit their safe environmental application (Köchermann et al., 2018; Yang et al., 2019). Recent studies have demonstrated that pH-controlled hydrothermal reactions can be tuned to produce artificial humic substances (A-HSs) that closely resemble natural humic matter in structure and functionality (Yang et al., 2019). These A-HSs, including artificial humic acids (A-HAs) and fulvic acids (A-FAs), are generated through the chemical transformation of biomass under hydrothermal conditions and offer a sustainable approach for soil regeneration and pollutant remediation (Yang et al., 2021). Utilizing these A-HAs in agroindustrial applications could represent one of the first large-scale products capable of achieving a carbon-negative footprint (Yang and Antonietti, 2020).

Building upon this concept, recently identified processes have expanded the hydrothermal framework by defining clear boundaries relative to conventional HTC. These modified hydrothermal pathways differ from HTC, which is conducted in the absence of alkaline agents under neutral to acidic conditions, by operating under controlled alkaline environments. In particular, hydrothermal humification (HTH) is performed under moderate alkalinity, whereas hydrothermal fulvicification (HTF) is carried out under strongly alkaline conditions, enabling a closer emulation of natural humification processes (Tkachenko et al., 2023). Under alkaline conditions, these processes promote the breakdown of lignocellulosic structures, generating artificial humic acids (A-HAs) and fulvic acids (A-FAs) (Ischia et al., 2026), instead of potentially harmful aromatics. By enhancing the quality of both solid and liquid fractions, HTH and HTF improve the environmental compatibility of hydrothermal products while aligning with circular bioeconomy principles (Marzban et al., 2024). Both processes are conducted under hydrothermal conditions comparable to HTC but differ primarily in the applied alkalinity. Moderate KOH addition in HTH favors artificial humic acid formation, whereas higher alkalinity in HTF enhances biomass solubilization and promotes fulvic-like compound generation, leading to a progressive shift from solid- to liquid-dominated products.

Humic substances (HSs) constitute up to 75 % of soil organic matter and play a critical role in nutrient cycling, soil fertility, and carbon sequestration (Ischia et al., 2026; Tkachenko et al., 2023). The unique condensed carbon structure of HS, enriched with oxygen-containing functional groups like carboxyl, carbonyl, phenolic hydroxyl, quinone, and methoxy, contributes to their exceptional physicochemical properties (Yang et al., 2021). Over long periods, natural environments foster the development of HSs. This process involves a combination of abiotic reactions and biological activities, with decomposition serving as a primary mechanism for their formation (Yang et al., 2019). In contrast, artificial humics (A-HS) from hydrothermal processes offer a faster, controllable, and renewable alternative (Yang et al., 2021). A-HS are gaining increasing interest because they represent the artificial counterparts of natural humic substances, sharing similar structural features and functional properties, and thus can mimic their roles in carbon cycling and nutrient retention. Beyond soil fertility, humic acids are increasingly applied in water treatment as adsorbents and complexing agents for pollutants, further broadening their utility (Sarlaki et al., 2024). Given these multifunctional properties, humic-based materials and their artificial analogs are particularly promising for integrated soil-water management systems.

Soil degradation and freshwater pollution are among the most pressing global environmental challenges, both intensified by population growth, industrialization, and unsustainable resource use. These challenges fall within critical planetary boundaries that have already been exceeded, indicating that Earth's capacity to sustain healthy soils and aquatic systems is under severe stress (Richardson et al., 2023). Industrial dyes are major contributors to aquatic pollution and are frequently used as model contaminants in adsorption-based remediation studies due to their persistence, toxicity, and well-defined physicochemical properties. Among them, Crystal Violet (CV) is a widely used cationic dye in textile, ink, and printing industries and is commonly detected in industrial effluents, where it can disrupt biological processes and pose significant risks to aquatic organisms and human health (Alizadeh et al., 2017). Owing to its environmental relevance and strong interaction with carbonaceous materials, CV is often employed as a representative probe molecule for evaluating adsorption performance. Adsorption is regarded as one of the most effective treatment strategies for dye-contaminated water, offering operational simplicity, cost efficiency, and high removal efficiency (Qadeer, 2007). In this context, hydrochar derived from renewable biomass sources has emerged as a promising low-cost adsorbent, enabling simultaneous pollutant removal and sustainable carbon valorization (Liu et al., 2021).

However, the safe and functional application of hydrochars requires not only high adsorption performance but also chemical

stability, particularly the absence of excessive leaching of soluble organic compounds that could compromise water quality or downstream applications. This aspect remains insufficiently addressed in many adsorption studies, where hydrochars are often used without systematic evaluation of leachate behavior. In addition, although HTC has been extensively studied, its alkaline-modified counterparts, HTH and HTF, have rarely been investigated within a unified framework, despite their profound influence on reaction pathways, carbon partitioning, and product functionality. Existing studies predominantly focus on the solid hydrochar fraction, even though increasing alkalinity intentionally redirects a substantial portion of biomass carbon into the liquid phase, which is commonly treated as a low-value byproduct despite its enrichment in humic- and fulvic-like compounds. The novelty of this study lies in providing the first systematic comparison of HTC, HTH, and HTF of grass biomass under identical hydrothermal conditions, with alkalinity as the sole governing parameter, while simultaneously evaluating the functional valorization of both solid and liquid products. Specifically, hydrochars are assessed in terms of leaching stability and adsorption performance toward a model cationic dye (Crystal Violet), whereas the corresponding process liquids enriched in artificial humic and fulvic substances are evaluated for bioactivity using seed germination assays. By explicitly linking hydrothermal reaction chemistry, carbon distribution, and end-use performance, this work demonstrates how controlled alkalinity enables a tunable hydrothermal platform that integrates water remediation and soil-related applications, moving beyond the traditionally fuel-oriented use of HTC hydrochars and advancing an integrated, circular carbon valorization strategy.

2. Materials and methods

2.1. Hydrochar preparation

Grass biomass was collected from a periodically flooded polder area in the Lower Oder Valley (LOV) National Park, Germany (53°02'32.6" N, 14°17'50.0" E). The feedstock had an initial moisture content of approximately 10 %, which was included in the total water balance, and distilled water was added to adjust the total solids (TS) content to 15 %. For hydrothermal carbonization (HTC), 88.9 g of grass and 444.5 g of water were used without alkali addition. For hydrothermal humification (HTH) and hydrothermal fulvification (HTF), the same amounts of grass was applied, with the addition 25.5 g and 50.9 g of KOH, respectively. The rationale for the selected KOH loadings for each process is described in the following section. The reaction mixture was placed in a 1 L sealed high-pressure batch reactor (Parr Instrument, Series 4520, Moline, Illinois, USA) and heated to 230 °C at a controlled heating rate of 2 K/min, with a residence time of 4 h measured from the point at which the reactor contents reached the target temperature of 230 °C. The reaction temperature was selected above the cellulose disintegration threshold to ensure effective conversion under HTC (Sevilla and Fuertes, 2009). The residence time was adopted from previous optimization studies in a 1 L reactor using different feedstocks (Marzban et al., 2024).

All experiments were performed under identical reaction conditions with respect to temperature, residence time, solid loading, and reactor configuration, and the only intentional variable distinguishing HTC, HTH, and HTF was the applied KOH concentration. The required alkali (KOH) was calculated based on the molar equivalent of the carbohydrate content in the biomass (Tkachenko et al., 2023). No alkali was added for hydrothermal carbonization (HTC), while 1.5 mol KOH per mol carbohydrate defined (33 wt.% of dry solid) the hydrothermal humification (HTH) condition, and 3 mol KOH per mol carbohydrate (66 wt.% of dry solid) corresponded to hydrothermal fulvification (HTF). These conditions are based on our previous works, in which we defined the boundaries between carbonization and humification and introduced the concept of hydrothermal fulvification (Tkachenko et al., 2023). We demonstrated that increasing alkaline loading systematically shifts hydrothermal reactions from carbonization (0–1 equivalent KOH per mol carbohydrate), where the added base is insufficient and the pH rapidly drops below 7, toward humification (1–2 equivalents KOH per mol carbohydrate), where the alkaline buffering capacity is adequate to prevent a sharp pH decrease and promotes auto-neutralization reactions (Tkachenko et al., 2023). At alkaline loadings above 3 equivalents, the process transitions to hydrothermal fulvification, in which biomass compartments, including carbohydrates, are extensively disintegrated, leading to a higher proportion and increased carbon transfer from biomass into liquid-phase product, as discussed in our previous work (Marzban et al., 2024). Table 1 summarizes the lignin, cellulose, and hemicellulose contents of the grass biomass used to calculate the required amount of KOH.

After completion of the reaction, the reactor was first allowed to cool naturally in the absence of heating until the temperature dropped below 100 °C, and was then immersed in water to achieve rapid cooling to room temperature. The reactor was subsequently opened, and the resulting slurry was separated into solid (hydrochar) and liquid phases by vacuum filtration using a 0.45 µm filter paper. The wet hydrochar was stored in sealed plastic bags, and the liquid fraction was kept at –18 °C until analysis. The hydrochar yield was calculated according to Eq. (1):

$$\text{Hydrochar yield (wt\%)} = \frac{\text{Dry mass of hydrochar (g)}}{\text{Dry mass of raw biomass (g)}} \times 100 \quad (1)$$

Artificial humic acid (A-HA) extraction from the liquid product followed the ISO 19822 protocol, with modifications by Marzban

Table 1
Fiber analysis of grass biomass.

NDF (%)	ADF (%)	ADL (%)	Lignin (%)	Hemicellulose (%)	Cellulose (%)
57.96 ± 1.53	30.72 ± 1.44	3.85 ± 1.17	3.85 ± 1.17	27.24 ± 2.10	26.87 ± 1.86

NDF: Neutral Detergent Fiber; ADF: Acid Detergent Fiber; ADL: Acid Detergent Lignin.

et al. (2024). The pH was adjusted to 1 with 6 M HCl, and the mixture was centrifuged at 4500 rpm for 5 min. In this study, humic- and fulvic-like fractions are distinguished operationally based on pH-dependent solubility, where the acid-precipitated fraction obtained after adjustment to pH 1 is referred to as artificial humic acid (A-HA), while the remaining acid-soluble fraction in the liquid phase is considered fulvic-like or dissolved humic matter. The separated A-HAs were filtered (Whatman), oven-dried at 60 °C for 48 h, and weighed to determine yield using Eq. (2):

$$A-HA \text{ yield (wt\%)} = \frac{\text{Dry mass of AHA (g)}}{\text{Dry mass of raw biomass (g)}} \times 100 \quad (2)$$

The hydrochars were labeled as GHTC, GHTH, and GHTF, corresponding to the hydrothermal carbonization, humification, and fulvification processes, respectively. The artificial humic acids obtained from HTH and HTF were labeled as A-HA-HTH and A-HA-HTF.

2.2. Analytical methods

The ANKOM A2000 Automated Fiber Analyzer was used to analyze the fiber composition of the raw materials, following the VDLUFA 2012 (Verband Deutscher Landwirtschaftlicher Untersuchungs- und Forschungsanstalten, 2013) methods. The analysis included measuring neutral detergent fiber (NDF), acid detergent fiber (ADF), and acid detergent lignin (ADL). Hemicellulose was calculated as the difference between NDF and ADF, and cellulose was determined by subtracting ADL from ADF. The lignin content was represented by the ADL values. These findings are presented in Table 1.

To determine the total solids (TS%) in hydrochar and A-HA, the samples were dried at 105 °C for 24 h. Following the DIN EN 14775 standard, the ash content was determined by heating the previously dried samples at 550 °C for 5 h. The organic total solids (OTS%) were calculated by deducting the ash content from the total solids measurement. A Vario ELIII elemental analyzer (Elementar Analysensysteme, Hanau, Germany) was used to measure carbon, hydrogen, nitrogen, and sulfur levels in the samples, using sulfonic acid as a reference. Prior to analysis, samples were dried at 60 °C. The oxygen content was calculated using the formula: O% = 100 – C% + H% + S% + N% + Ash%). In fuel-oriented applications of hydrochar, an increased higher heating value (HHV) enhances combustion efficiency while simultaneously lowering transportation expenses. According to the findings of Marzban et al. (2022), a relationship can be established between the elemental composition of hydrochar and its experimentally determined HHV (Eq. (3)). Accordingly, the Eq. (3) proposed in their study was employed to estimate the HHV for all hydrochar samples in the present work.

$$HHV_{HC} = (0.3853 \times C) + \left(\frac{44.98}{O}\right) \quad (3)$$

where O (oxygen, wt.%) and C (carbon, wt.%) are in wt.% and HHV_{HC} is in MJ/kg.

Sugars (glucose, sucrose, and fructose), aromatics (phenol, cresol, catechol, guaiacol, furfural, and hydroxymethylfurfural (HMF)), and organic acids (acetic, lactic, and formic) were analyzed as the main organic compounds found in both the hydrochar and the liquid product. Aromatics were identified using a Thermo Fisher ICS-3000 HPLC system equipped with a UV detector, while the inorganic content was analyzed through ICP-OES (Thermo Scientific iCAP6300 Duo) coupled with a CETAC ASX-520 autosampler. A Dionex Ultimate 3000 UPLC system, equipped with a KNAUER Eurokat H column (300 mm×8 mm, 10 μm), was used to measure sugars and organic acids (Marzban et al., 2023). For the liquid product, additional parameters such as pH, total carbon (TC), and total organic carbon (TOC) were measured using a Shimadzu TOC-5050A Analyzer. Phosphate, nitrate, nitrite, and ammonium concentrations in both liquid product and hydrochar samples were determined using a Thermo Fisher ICS 1000 ion chromatograph.

The surface morphology of all hydrochars samples from HTC, HTH and HTF processes and the original feedstock, was investigated using Scanning Electron Microscopy (Phenom ProX Desktop SEM, USA). The Brunauer-Emmett-Teller (BET) method was used to evaluate specific surface area, pore size, total pore volume, and CO₂ adsorption/desorption isotherms, utilizing a Micromeritics TriStar II analyzer (USA). For functional group identification, Fourier Transform Infrared Spectroscopy (FTIR) was carried out using a Nicolet iS5 ATR-FTIR spectrometer (Thermo Scientific, Madison, WI, USA). Thermogravimetric Analysis (TGA) was utilized to assess the thermal stability of all hydrochars. The samples were first maintained at 100 °C for 10 min to eliminate moisture, followed by a controlled heating process up to 1000 °C at a rate of 10 °C min⁻¹, from which the T₅₀ value, the temperature corresponding to 50 % weight loss, was subsequently determined. The particle size and zeta potential measurements of solutions were carried out using a Zetasezer Nano Ultra equipped with a backscattering detector (Klučáková and Kalina, 2015). Excitation Emission fluorescence spectra (EEM) were recorded with a Perkin-Elmer LS50B fluorescence spectrophotometer, using excitation wavelengths ranging from 200 to 450 nm in 5 nm steps and emission wavelengths from 250 to 600 nm in 0.5 nm steps. Both excitation and emission slit widths were set to 10 nm, and the scanning speed was maintained at 1200 nm/min.

2.3. Hydrochar washing process

A sequential washing procedure was conducted to evaluate the potential release of soluble components from hydrochar upon contact with water, simulating conditions relevant to soil application. After the HTH and HTF processes, the resulting hydrochar was divided into two batches to assess the effect of drying on leaching behavior. One batch was oven-dried at 105 °C for 24 h and stored in a desiccator, while the other was stored immediately after filtration in airtight containers to preserve its original moisture content. Both dried and wet hydrochars were subjected to sequential washing to evaluate the release of water-soluble components. For each cycle, 1 g of hydrochar (dry weight basis) was mixed with 40 mL of deionized water and shaken at 150 rpm for 1 h at room temperature. The

suspension was centrifuged at 4500 rpm for 15 min, and the remained liquid was collected as leachate. Fresh deionized water (40 mL) was then added to the residue, and washing was repeated up to 18 cycles until no coloration or detectable organics were observed.

2.4. Adsorption study

A 500 mg/L crystal violet (CV) stock solution was prepared by dissolving 0.5 g of dye in 1 L of distilled water using a volumetric flask and stored in the dark to prevent photodegradation. The stock solution was diluted to obtain the desired CV concentrations for adsorption studies. Batch experiments were conducted by mixing 50 mg of hydrochar with 50 mL of dye solution in conical tubes, shaken at 25 °C and 200 rpm for a contact time of 3 h. Initially, three hydrochars (HTC, HTH, and HTF) were screened for CV removal efficiency, and the most effective one was selected for further investigation. After adsorption, samples were separated using paper filtration, and the residual CV concentration was determined by UV-Vis spectroscopy at a maximum absorption wavelength ($\lambda_{\max}$) of 590 nm. All adsorption experiments were conducted in duplicate, and mean values with standard deviations were reported. Subsequent experiments examined the influence of key parameters, initial CV concentration (10–500 mg/L), solution pH (3–11), and adsorbent dosage (0.1–10 g/L), on adsorption performance. The equilibrium time was determined at varying initial concentrations while maintaining a constant adsorbent loading of 1 g/L and fixed solution pH. The adsorption capacity and removal efficiency of adsorbent was determined as follow (Eq. (4), Eq. (5)):

$$q_e = \frac{(C_0 - C_e) * V}{W} \quad (4)$$

$$\text{Removal efficiency}\% = \frac{C_0 - C_e}{C_0} \times 100 \quad (5)$$

where q_e is represents the quantity of adsorbate adsorbed per unit mass of adsorbent (mg/g), V is the volume of solution (L), C_0 is the initial concentration of the adsorbate (mg/L), C_e is the equilibrium concentration of the adsorbate (mg/L) and W is the mass of adsorbent (g).

The adsorption performance was analyzed by evaluating equilibrium isotherms and kinetic data, utilizing the selected hydrochar that was obtained from the HTC, HTH and HTF processes. To investigate the adsorption behavior between the adsorbent and adsorbate, experimental data were fitted to four isotherm models: Langmuir (Eq. (6)), Freundlich (Eq. (7)), Sips (Eq. (8)) and Redlich–Peterson (Eq. (9)). The Langmuir model describes adsorption occurring in a single molecular layer on a surface with identical and energetically equivalent sites, under the assumption that a dynamic equilibrium exists between the rates of adsorption and desorption (Ayawei et al., 2017). The Freundlich isotherm, on the other hand, is suited for adsorption on heterogeneous surfaces with varying affinities. Sips model suggested the combination of the Langmuir and Freundlich, while Redlich–Peterson proposed a three-parameter model which is not representing an ideal monolayer adsorption (Kalam et al., 2021).

$$q_e = \frac{Q_m K_L C_e}{1 + K_L C_e} \quad (6)$$

$$q_e = K_F C_e^{\frac{1}{n}} \quad (7)$$

$$q_e = \frac{q_s K_s C_e^{\beta_s}}{1 + K_s C_e^{\beta_s}} \quad (8)$$

$$q_e = \frac{K_r C_e}{1 + \alpha C_e^{\beta}} \quad (9)$$

where q_e is the adsorption capacity of the adsorbent (mg/g) at equilibrium condition, Q_m is the maximum adsorption capacity (mg/g), K_L is the Langmuir constant (L/mg), K_F is the distribution coefficient (L/mg), C_e is the CV concentration at equilibrium (mg/L), n is the Freundlich constant, β_s is model exponent, q_s maximum adsorption capacity (mg/g) and K_s (L/mg) is Sips model constant, K_r (L/g) and α (L/mg) are Redlich–Peterson's constant, and β is an exponent that range from 0 to 1. The adsorption kinetics of CV onto the surface of the selected hydrochar was similarly explored by fitting the experimental data to pseudo-first order, pseudo-second order and Weber–Morris diffusion kinetic models (Eq. (10), Eq. (11) and Eq. (12)) for analysis (Ho and Mckay, 1998).

$$q_t = q_e(1 - e^{-k_1 t}) \quad (10)$$

$$q_t = \frac{k_2 q_e^2 t}{1 + K_2 q_e t} \quad (11)$$

$$q_t = k_{id} t^{1/2} + C \quad (12)$$

where q_t is the adsorption capacity of the adsorbent (mg/g) at time t , k_1 is the pseudo-first order rate constant (1/min), k_2 is the pseudo-second order constant (g/min.mg), k_{id} is intraparticle diffusion rate content (mg/g.min^{1/2}) and C is the boundary layer thickness

constant.

2.5. Germination assay

A germination experiment was carried out to assess the influence of process liquids of HTC, HTH and HTF on the reearly seedling development of spring oilseed rape seed (*Brassica napus* - variety "Lakritz"). Plastic trays were filled with Einheitserde - CL Pikier growing substrate, into which the prepared liquids were thoroughly incorporated at different dilution rates of 0, 10, 25 and 50 times. Each treatment was replicated three times, and the mean values have been reported. Prior to assessment, ten seeds were planted in each tray, and both the germination percentage (GP) and seed vigor index (SVI) were measured (Eq. (13), Eq. (14)) (Jogaiah, 2014):

$$\text{Germination Percentage (GP)} = \text{SNG}/\text{SNO} \times 100 \quad (13)$$

$$\text{Seed Vigour Index (SVI)} = \text{GP} \times \text{Seeding length} \quad (14)$$

where SNG is the number of germinated seeds and SNO represents the total number of viable seeds.

3. Results and discussion

3.1. Elemental analysis of grass products from hydrothermal processes

The yields and elemental compositions of the, rawgrass, hydrochars (GHTC, GHTH, GHTF) and extracted artificial humic acids (A-HAH from HTH and A-HAF from HTF) are summarized in Table 2. Solid yields decreased from 54.1 % in the carbonization process (GHTC) to 44.5 % in the humification stage (GHTH) and 13.8 % in the fulvification stage (GHTF), reflecting enhanced solubilization of organic matter under alkaline conditions at 230 °C. A decrease in solid yield with increasing hydrothermal severity has also been reported in previous HTC studies (Petrovic et al., 2021), where enhanced depolymerization and solubilization of biomass under more severe conditions led to a greater transfer of carbon into the liquid phase. The GHTC sample showed the lowest ash content and an acidic pH of 4.26, while the addition of KOH in the humification and fulvification treatments led to inorganic accumulation and increased pH values of 8.05 and 9.39, respectively (Tkachenko et al., 2023). Alkali addition also markedly enhanced the formation of artificial humic acids, with yields rising from nearly zero in HTC to 2.25 % in HTH and 8.8 % in HTF. The C content of GHTC reached 60.6 %, exceeding that of the raw grass feedstock (42.1 %) and the alkaline-assisted samples (42.3 % and 31.7 % for GHTH and GHTF, respectively). This enrichment, along with lower ash content, resulted in a substantially higher heating value (25.70 MJ/kg) (Marzban et al., 2022).

The H/C and O/C atomic ratios, shown in Table 2 and illustrated in the Van Krevelen diagram (Fig. 1), provide insight into the chemical structure and carbonization degree of the samples. In addition to compositional information, these ratios are also indicative of the degree of carbonization and overall chemical stability of the materials, with lower H/C and O/C values generally corresponding to more condensed and stable carbon structures (Koprivica et al., 2024). The carbonization step, dominated by dehydration reactions, produced a hydrochar with higher carbon content than the original grass biomass. In contrast, humified and fulvified samples (GHTH and GHTF) showed progressively higher H/C and O/C ratios, reflecting increased oxidation and incorporation of oxygenated functional groups. Such systematic shifts in H/C and O/C ratios with increasing hydrothermal severity are well documented in HTC literature and are attributed to progressive dehydration, decarboxylation, and aromatization reactions during carbon transformation (Petrovic et al., 2021). The relatively high H/C ratio in GHTF likely resulted from decarboxylation, indicating the formation of hydrogen-rich, fulvic-like structures consistent with the transformations reported for hydrothermal fulvification (Kozyatnyk et al., 2015). Similar trends were observed in the A-HAs extracted from the liquid phase, with H/C and O/C ratios of approximately 1.39 and 0.24 for A-HAF, and 1.30 and 0.45 for A-HAH, respectively, indicating hydrocarbon- and lignin-derived aromatic structures. Both ratios are consistent with values reported for artificial humic substances in previous studies (Marzban et al., 2024), confirming the reproducibility of these chemical features.

Nitrogen (N) content exhibited a similar pattern to that of carbon, progressively decreasing from hydrothermal carbonization through humification to fulvification. The C/N ratio increased steadily across the treatments, from 38.2 in GHTC to 57.6 in GHTH and reaching 79.0 in GHTF. This upward trend indicates a gradual enrichment in C relative to N, suggesting that the organic matter likely becomes more chemically stable and less nitrogen-rich as it progresses from carbonization to humification and ultimately to

Table 2

Elemental analysis, yield, pH, and higher heating value (HHV) of hydrochars and extracted A-HAs (The mean values are shown; corresponding standard deviations are provided in the Table SII of Supplementary Information).

Sample	Ash (%)	C (%)	H (%)	N (%)	S (%)	O (%)	C/N	H/C	O/C	pH	Yield (%)	HHV _{HC} (MJ/kg)
Grass	9.45	42.13	6.93	1.27	0.23	39.99	33.12	1.97	0.71	-	-	17.36
GHTC	13.14	60.60	4.98	1.59	0.26	19.42	38.23	0.99	0.24	4.26	54.1	25.70
GHTH	23.73	42.30	5.55	0.74	0.17	27.51	57.53	1.57	0.49	8.05	44.5	17.93
GHTF	42.14	31.70	4.99	0.40	0.13	20.64	79.00	1.89	0.49	9.39	13.18	14.40
A-HAH	29.51	40.38	4.37	1.47	0.22	24.06	27.45	1.30	0.45	-	2.25	-
A-HAF	32.09	46.60	5.40	1.01	0.12	14.78	46.08	1.39	0.24	-	8.8	-

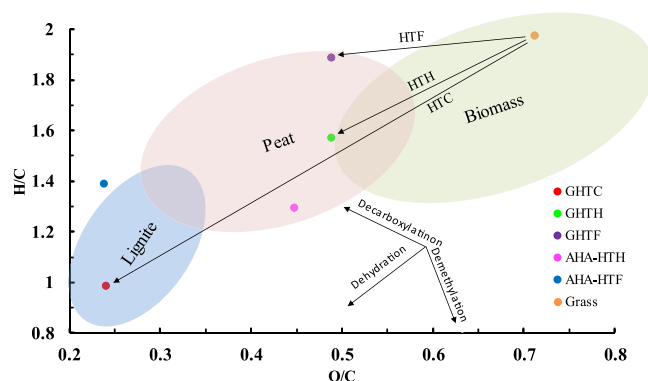


Fig. 1. Van Krevelen diagram of grass, hydrochars and extracted A-HA from hydrothermal carbonization, humification and fulvification processes.

fulvification. Such a pattern reflects reduced biodegradability and an increasing degree of molecular condensation, particularly under fulvification conditions (Celletti et al., 2021). The C/N ratio of the artificial humic acid (A-HA) was 27.5, slightly above the critical threshold of 24:1 (Melillo et al., 1982), indicating a potential for temporary N immobilization in soil. However, this effect is likely occurring in mild and short-term, with gradual nitrogen release over time.

3.2. Distribution of organics and nutrients in hydrothermal processes

Fig. 2-a illustrates the variation in total carbon (TC) and total organic carbon (TOC) concentrations in the liquid phase across the HTC, HTH, and HTF processes. The mean values and standard deviations of TC, TOC, organic acids, aromatics, sugars, and ions are summarized in Table SI2. The HTF treatment yielded markedly higher TC and TOC levels than HTC and HTH, reaching 62.15 and 58.53 g/L, respectively. These results confirm that most of the carbon in the liquid phase is organic in nature. Notably, the TOC concentration under HTF was approximately 3.7 times higher than that observed in HTC, demonstrating the superior efficiency of the fulvification process in mobilizing organic carbon into the aqueous phase.

In the HTC condition, lactic, formic, and acetic acid levels in the liquid phase were relatively lower than HTH and HTF, recorded at 1.84, 0.64, and 6.90 g/L, respectively (Fig. 2-b). These modest concentrations suggest limited biomass breakdown under the investigated operating reaction conditions. Correspondingly, the hydrochar contained only small amounts of lactic (1.03 g/kg DM) and acetic acid (5.20 g/kg DM), while formic acid was not detected, indicating minimal incorporation of these products into the solid phase. Under HTH, acid concentrations increased markedly in both liquid and solid fractions. In the liquid, lactic acid rose to 23.54 g/L, and formic acid to 8.33 g/L. Acetic acid remained relatively stable at 7.70 g/L, suggesting that its formation may plateau under moderate alkalinity. The solid fraction showed a similar trend, with lactic acid increasing to 19.71 g/kg DM, and formic and acetic acid to 7.07 and 5.93 g/kg DM, respectively. Further enhancement occurred under HTF, where the liquid phase showed the highest concentrations of all three acids. Lactic acid peaked at 39.17 g/L, almost twenty times more than the level observed under HTC. Formic

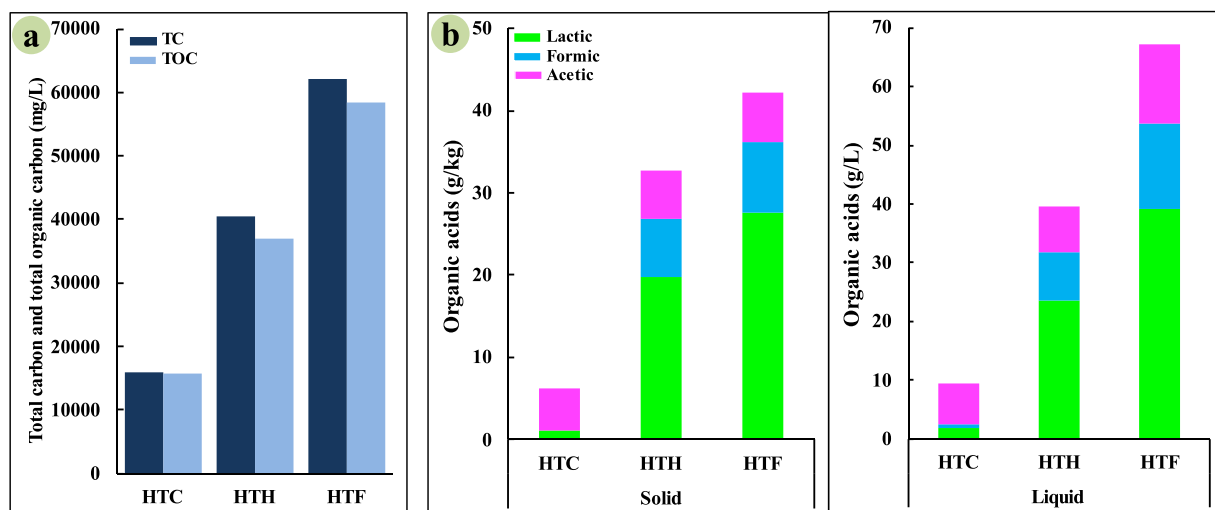


Fig. 2. The concentration of a) total carbon and total organic carbon in liquids (mg/L), b) organic acids in both solids and liquids from different processes.

acid rose sharply to 14.45 g/L, and acetic acid more than doubled compared to HTC, reaching 13.60 g/L. These sharp increases indicate intensified biomass decomposition and secondary reactions under HTF treatment, likely driven by increased KOH content, facilitating acid-forming pathways. The solid phase under HTF also showed elevated acid contents (lactic 27.61, formic 8.66, and acetic 5.88 g/kg DM).

Fig. 3 illustrates the distribution of sugars and aromatics in both the solid and the liquid products from the hydrothermal treatments. Glucose was the dominant sugar in the humification and fulvification stages, showing a clear increase from 0.07 g/kg in hydrothermal carbonization (HTC) to 0.25 g/kg in hydrothermal humification (HTH) and 6.93 g/kg in hydrothermal fulvification (HTF) (Fig. 3-a). In contrast, the glucose concentration in the liquid phase rose slightly from 0.09 g/L in HTC to 0.19 g/L in HTH but was undetectable in HTF. This inverse trend between the solid and the liquid phases indicates progressive redistribution and transformation of glucose as processing severity and alkalinity increase. The disappearance of glucose from the liquid fraction under HTF likely reflects its degradation into carboxylic acids, furans, or aromatic intermediates under stronger alkali conditions (Ischia et al., 2026; Marzban et al., 2024). Sucrose was detected only in HTC (1.75 g/kg in the solid and 0.46 g/L in the liquid). Fructose appeared solely in the liquid fraction of HTH (0.02 g/L), possibly as a transient intermediate from sucrose hydrolysis or glucose isomerization. Its absence in HTC and HTF highlights its instability and high reactivity under hydrothermal conditions, particularly at the greater alkalinity and severity characteristic of HTF.

In the HTC condition, a broad range of aromatic compounds was detected in both liquid and solid phases (Fig. 3-b). HMF reached 161.23 mg/L in the liquid and 172.12 mg/kg DM in the hydrochar. Catechol and phenol were also abundant, with liquid

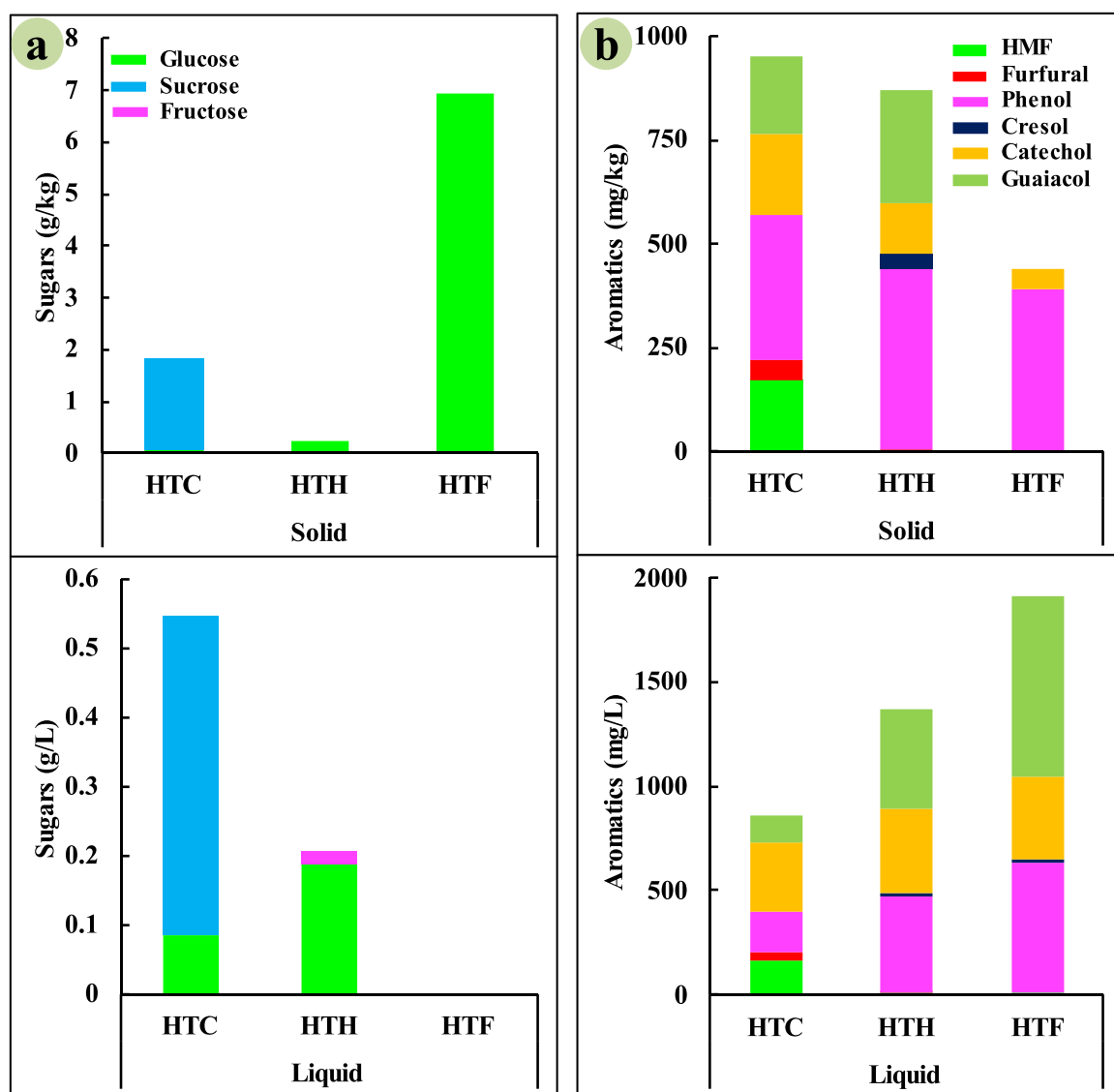


Fig. 3. Concentrations of a) sugars content and b) aromatic compounds in hydrochars (mg/kg) and process liquids (mg/L).

concentrations of 334.24 and 195.66 mg/L, respectively. In the solid, catechol and phenol reached 195.64 and 347.82 mg/kg DM, while guaiacol measured 126.64 mg/L in the liquid and 188.16 mg/kg DM in the solid. Furfural was present at 39.21 mg/L in the liquid and 49.96 mg/kg DM in the solid, whereas cresol was undetected. Under HTH, the liquid concentrations of HMF and furfural declined sharply to 7.40 and 3.68 mg/L, respectively. In contrast, phenol nearly doubled to 462.49 mg/L, with 434.59 mg/kg DM in the solid. Catechol rose to 399.52 mg/L in the liquid but dropped to 123.97 mg/kg DM in the solid, while guaiacol increased markedly to 475.70 mg/L and 271.34 mg/kg DM. Cresol appeared for the first time, reaching 17.64 mg/L in the liquid and 37.16 mg/kg DM in the solid. Under HTF, aromatic production further intensified: phenol peaked at 623.55 mg/L, catechol at 393.81 mg/L, and guaiacol at 863.18 mg/L. Cresol rose slightly to 20.07 mg/L, while HMF and furfural were nearly absent. In the solid, phenol remained high at 389.80 mg/kg DM, though catechol dropped to 50.75 mg/kg DM, and guaiacol and cresol disappeared.

Data provide some insights into the chemical pathways governing the conversion. Under acidic conditions (HTC), acetic acid is the dominant product, deriving from the dehydration pathway of glucose through fructose and HMF, followed by oxidation and rehydration steps that lead to acetate. This is a common path typically observed in HTC (Nicolae et al., 2020). Its high presence on the solid phase is likely due to its adsorption on the hydrochar surface. Instead, lactic acid remains minor, since its formation via retro-aldol cleavage of aldehydic intermediates is disfavored under acidic conditions. Consistently, HMF is detected in both liquid and solid phases, as it forms directly from glucose and cellulose/hemicellulose dehydration and undergoes subsequent back repolymerization into polyfuranic domains of the solid phase (Nicolae et al., 2020). Furfural follows the same route from pentoses. In contrast, as the reaction environment shifts to alkaline HTH and HTF conditions, the furanic route is suppressed: HMF disappears, while the retro-aldol

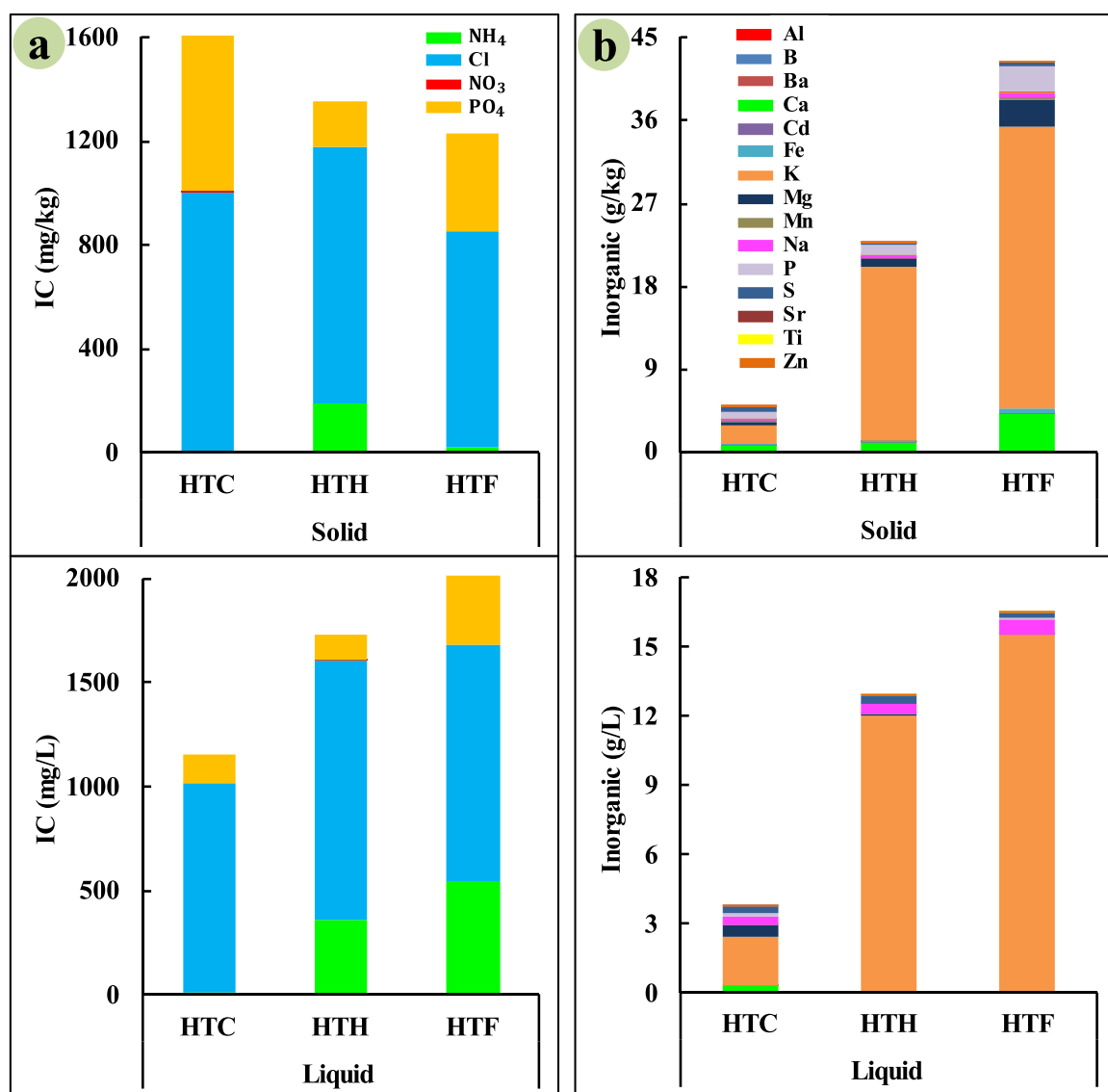


Fig. 4. The concentration of ions measured by IC (a) and inorganics (b) in the liquids (mg/L) and solids (mg/kg).

pathway of carbohydrates becomes dominant, producing lactic, formic, and acetic acids. At the same time, phenolic derivatives strongly increase. Although they are classically linked to lignin depolymerization (β -O-4 cleavage and solubilization), the low lignin content here (3.85 %) suggests that a substantial fraction may also originate from carbohydrate-derived intermediates. Indeed, retro-aldol products can undergo base-catalyzed aldol condensation, cyclization, and partial aromatization, yielding phenolic-like structures (Kruse and Dinjus, 2005). This carbohydrate-to-phenolic route has been observed in other alkaline hydrothermal systems (Ischia et al., 2026) and still remains an open question for further studies. Then, increasing alkalinity further enhances biomass decomposition, explaining the progressive transfer of phenolic species from solid to liquid phase. The higher abundance of phenol compared to guaiacol and other methoxy-phenols may reflect demethoxylation under strong alkaline conditions or the contribution of non-lignin pathways (Toor et al., 2011). Overall, the data support the idea that under HTH and HTF, the phenolic fraction results from both lignin solubilization and secondary reactions of carbohydrate intermediates. Therefore, taken together, these observations indicate that increasing alkalinity not only promotes carbohydrate depolymerization and lignin solubilization, but also redirects the resulting intermediates, sugars, aldehydes, and furans, toward small organic acids and phenolic motifs. The conversion pathway agrees with base-catalyzed retro-aldol and condensation mechanisms as alkalinity increases. The shift leads to the formation of precursors like aldehydic intermediates and lignin fragments, which can undergo polycondensation and/or aromatization further forming to humic-like structures. Overall, it underlines the role of alkaline hydrothermal conditions in promoting the synthesis of A-HS.

The distribution of ionic species and trace elements in both liquid and solid phases following hydrothermal processing is shown in Fig. 4-a and b, respectively. In the absence of alkaline (HTC), the concentration of nutrients such as NH_4^+ and PO_4^{3-} was relatively low,

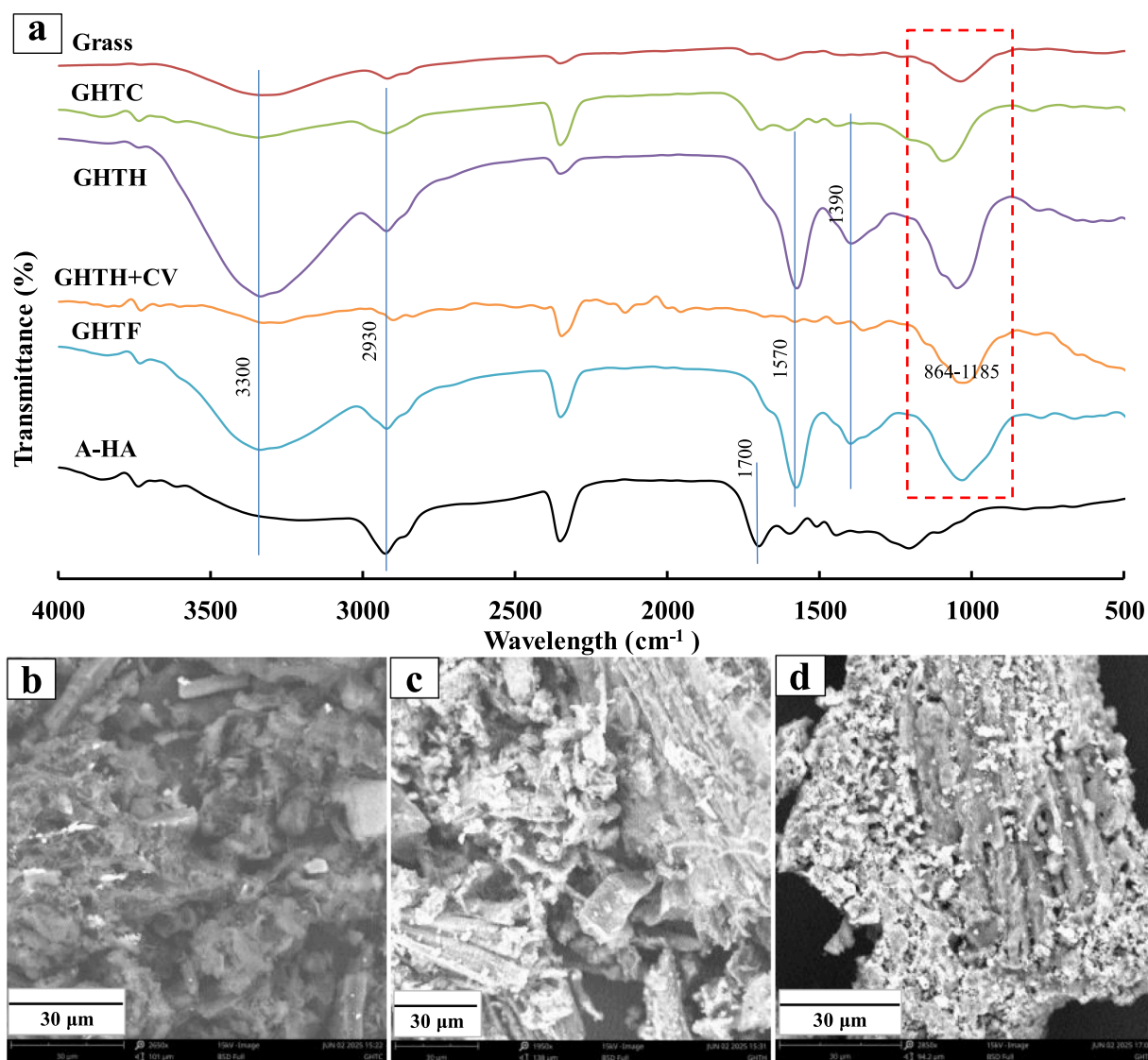


Fig. 5. FT-IR spectra of grass, hydrochars from HTC, HTH, HTF and A-HA (a), SEM analysis of HTC hydrochar (b), HTH hydrochar (c), and HTF hydrochar (d).

with NH_4^+ at 12.37 mg/L and PO_4^{3-} at 136.59 mg/L in the liquid phase. In contrast, under HTF conditions, these values increased substantially to 547.11 and 380.92 mg/L, respectively. Solid-phase phosphorus concentrations followed an inverse pattern, decreasing from 619 mg/kg DM in HTC to 378.8 mg/kg DM in HTF. Notably, chloride concentrations were consistently high across all conditions, with a peak of 2151.10 mg/L under HTC in the liquid phase. The total ammonium and phosphate concentrations in the liquid phase increased with the transition from HTC to HTF, suggesting enhanced solubilization of these ions.

The HTF condition resulted in a pronounced increase in calcium (4.13 g/kg), magnesium (2.94 g/kg), and potassium (30.65 g/kg) in the solid phase, surpassing levels found under HTC and HTH. Simultaneously, metals such as Fe, Mn, and Zn showed substantial increases in solid-phase concentrations, particularly Fe (0.44 g/kg) and Mn (0.26 g/kg). This pronounced accumulation of inorganic nutrients and trace metals in the HTF-derived solid directly explains its elevated ash content (42.14 %), reflecting mineral enrichment during fulvification and supporting its potential value as a nutrient-bearing soil amendment rather than a fuel material. In the liquid phase, phosphorus and sodium were the highest under HTF, at 0.05 g/L and 0.65 g/L, respectively. Several elements, including Co, Pb, Sb, Se, and Sn, remained below the detection limits under most of the conditions. Overall, the HTF process was the most effective at mobilizing both macro- and micronutrients into the solid and liquid fractions, demonstrating its potential for nutrient recovery and resource utilization.

3.3. Characterization of hydrochar and artificial humic acid

The FTIR spectra (Fig. 5-a) demonstrate the progressive transformation of grass biomass through hydrothermal treatments and artificial humification. The O–H stretching band (3300 cm^{-1}) became more intense and broader from GHTC to GHTF, suggesting increased hydroxyl exposure and overlapping phenolic or carboxylic groups, while its disappearance in A-HA, indicating the loss of hydrogen-bonded hydroxyls during synthesis. Aliphatic C–H bands ($2930\text{--}2850\text{ cm}^{-1}$) became sharper in GHTH and GHTF, reflecting thermal cleavage and formation of aliphatic structures, while a distinct C=O peak at 1700 cm^{-1} in A-HA signals oxidative processes. The $1570\text{--}1390\text{ cm}^{-1}$ region revealed aromatic and carboxylate vibrations typical of humic substances (Cunha et al., 2009). Strong absorptions between 1185 and 864 cm^{-1} in GHTH and GHTF, marked in the red box, indicated increased polysaccharide- and ether-derived C–O functionalities, consistent with lignocellulose degradation and oxygenated groups enrichment (Pandey, 1999), during humification (Marzban et al., 2024). Comparable FTIR characteristics between artificial humic acids and reference humic substances have been reported for alkaline-assisted hydrothermal systems, supporting the interpretation of A-HA as chemically and functionally humic-like materials (Ghaslani et al., 2024).

In addition, comparison of GHTH before and after crystal violet adsorption (GHTH+CV) reveals clear spectral changes associated with oxygen-containing functional groups, including variations in the O–H stretching region (3300 cm^{-1}), the C=O stretching region (1700 cm^{-1}), and the C–O stretching region between 1185 and 864 cm^{-1} . Notably, the band around 1570 cm^{-1} , commonly attributed to aromatic C=C or asymmetric carboxylate vibrations, is significantly reduced or disappears after CV adsorption. These spectral modifications indicate the involvement of carboxylate and phenolic functional groups in CV binding, providing spectroscopic evidence for interaction between GHTH and the cationic dye (Shoukat et al., 2017).

Fig. 5-b, c, d present SEM images of hydrochars produced under different hydrothermal treatments, revealing distinct morphological transformations. GHTC (Fig. 5-b) exhibited a compact, smooth surface with moderate porosity, typical of materials formed primarily through dehydration and condensation reactions. In contrast, GHTH (Fig. 5-c), and GHTF (Fig. 5-d) displayed increased shrunken and fragmented structures, indicating enhanced conversion of carbon from the solid to the liquid phase. This observation is supported by significantly higher total organic carbon (TOC) concentrations in the liquid phase of GHTF (58.53 g/L), compared to GHTH and GHTC, which showed markedly lower TOC values, as highlighted in Fig. 2-a. The increase in liquid-phase TOC is accompanied by a sharp decline in solid yield, dropping from 54.1 % in GHTC to 44.5 % in GHTH, and reaching only 13.2 % in GHTF, as detailed in Table 2. These results highlight the progressive solubilization and degradation of biomass during the fulvification process.

Fig. 6 shows the TGA results, which reveal clear differences in thermal reactivity among the hydrochars depending on KOH treatment. Grass shows the lowest stability ($T_{50} = 319^\circ\text{C}$; ash = 11.26 %) due to the decomposition of untreated lignocellulose. GHTC,

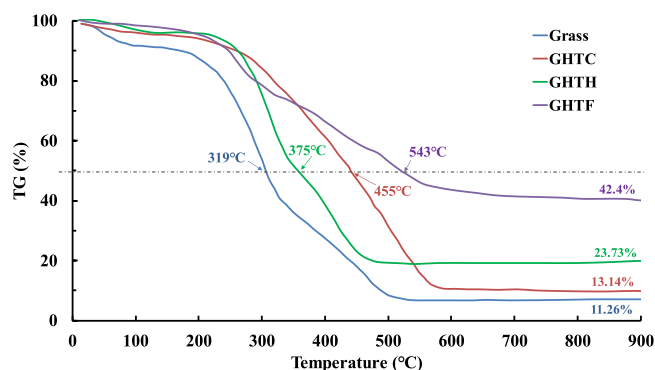


Fig. 6. Thermogravimetric analysis (TGA) of grass and hydrochars obtained from HTC, HTH, and HTF processes.

produced without KOH, displays a markedly higher T_{50} of 455 °C but leaves only 13.14 % ash, consistent with dehydration and condensation reactions that enhance structural ordering and thermal resistance (Sevilla and Fuertes, 2009). When KOH is introduced into GHTH, the T_{50} decreases to 375 °C, while the ash increases to 23.73 %, suggesting that alkali catalysis promotes mineral retention and partial stabilization of the carbon matrix despite lowering the onset of thermal degradation. With double KOH loading in GHTF, thermal resistance is maximized, with the highest T_{50} (543 °C) and largest residue (42.4 %), reflecting alkali-driven carbon condensation and incorporation of inorganic phases that contribute to char stabilization (Zhu et al., 2015). To account for the influence of inorganic residues (ash%), the T_{50} values were further corrected on an ash-free basis, as outlined in the methodology reported by Harvey et al. (2012). After correction, the T_{50} values were 312 °C for grass, 435 °C for GHTC, 332 °C for GHTH, and 389 °C for GHTF (Fig. S11). These results reveal a shift in the relative thermal stability ranking, with GHTC maintaining the highest stability among the samples, while the apparent advantage of GHTF was reduced once mineral contributions are excluded. This adjustment highlights that part of the stability observed in the uncorrected data, particularly for GHTF, is attributable to inorganic phase retention rather than intrinsic carbon matrix stabilization. Overall, the ash-corrected T_{50} values indicate that although alkali treatment facilitates carbon condensation, the observed enhancement in thermal stability, especially in GHTF, is largely due to inorganic residue retention rather than a true increase in the intrinsic thermal resistance of the carbon matrix.

BET analysis (Table 3) reveals distinct differences in textural properties among the prepared adsorbents. GHTC exhibited a relatively high surface area of 45.50 m²/g, attributed to the partial carbonization and development of micropores during dehydration and decarboxylation, with a pore size of 2.33 nm and a total pore volume of 0.0266 cm³/g. In contrast, GHTH showed a lower surface area of 18.59 m²/g, with a similar pore size of 2.31 nm and a reduced pore volume of 0.0107 cm³/g. This is consistent with the (Fig. 1), Van Krevelen diagram, which indicates increased oxygenation and incorporation of polar functional groups during humification. Although this reduces porosity, it enriches the surface with oxygenated functional groups that enhance adsorption properties. GHTF showed the highest surface area (51.56 m²/g), the largest pore size (5.15 nm), and the highest pore volume (0.0663 cm³/g), suggesting that fulvification could promote the formation of mesopores and a more open pore network. Overall, these results highlight that surface chemistry and porosity evolve differently across the processes, with HTH favoring functionalization and HTF combining both high surface area and abundant active sites.

The particle size distribution data (Fig. 7a-c) show that GHTF products contain a significant higher proportion of smaller particles than GHTH, reflecting a finer and less aggregated colloidal structure consistent with previous reports (Esfahani et al., 2015). This is supported by zeta potential analysis, which showed that GHTF displayed a markedly more negative value (-8.72 ± 1.71 mV) than GHTH (-1.61 ± 0.02 mV), suggesting higher electronical repulsion and therefore enhanced colloidal stability in the fulvification process liquid. Conductivity measurements further differentiate the two, with GHTH showing higher conductivity (1.09 ± 0.01 mS/cm) than GHTF (0.58 ± 0.01 mS/cm), pointing to a greater concentration of free ionic species dissolved in the humification liquid. Together, these results indicate that fulvification produces smaller and more stable colloids with higher negative surface charge, while humification retains more dissolved ions in solution, which lead to higher aggregation. This specific behavior might be attributed to the humic substances present in both hydrothermal process liquids. Fluorescence spectra recorded on the two processes liquids (Fig. 7d-e) clearly demonstrate the presence of peaks raising in the region specifically associated to humic substances (Ex 320–360 nm / Em 420–480 nm) (Matthews et al., 1996), which intensity and shape became sharper and better defined after acidic components extraction. After extraction, these peaks become sharper and more defined, accompanied by a noticeable shift of the main fluorescence region toward higher excitation and emission wavelengths, further supporting the enrichment of humic-like components.

3.4. Hydrochar washing

The sequential washing of hydrochars over 18 cycles revealed distinct patterns of carbon-leachate behaviors associated with each hydrothermal treatment. Both total carbon (TC) and total organic carbon (TOC) showed a steep decline after the first few washing cycles (Fig. 8) indicating that most of the leachable carbonaceous content was released during the early stages. Hydrochar produced under HTH conditions exhibited the highest initial TC and TOC concentrations (155 g/kg), reflecting the presence of water-soluble humic substances formed through partial humification. In contrast, the HTC-produced hydrochar showed comparatively low TC and TOC values, confirming that the process promotes condensation and stabilization of organic matter into less soluble structures, with TOC concentrations in the leachates falling below the detection limit of 1 mg/L for the TOC analyzer (Shimadzu TOC-5050A) after the final washing cycles. This finding is consistent with the lower H/C and O/C ratios observed for HTC products, which indicate advanced carbonization and greater chemical stability (Kozyatnyk et al., 2023).

The HTF process also generated high initial TC and TOC concentrations (130 g/kg), but these values decreased more gradually across subsequent washing cycles, demonstrating the persistent solubility of fulvic-like compounds (Tkachenko et al., 2023). Interestingly, omission of the drying step before washing (HTF-no-drying) reduced the initial carbon release to 90 g/kg, suggesting that drying promotes mild oxidation or structural rearrangement that increases the leachability of organic components. Overall, the

Table 3

Specific surface area, pore size and total pore volume of different hydrochars from hydrothermal processes.

	GHTC	GHTH	GHTF
Specific surface area (m ² /g)	45.50	18.59	51.56
Pore size (nm)	2.33	2.31	5.15
Total pore volume (cm ³ /g)	0.0266	0.0107	0.0663

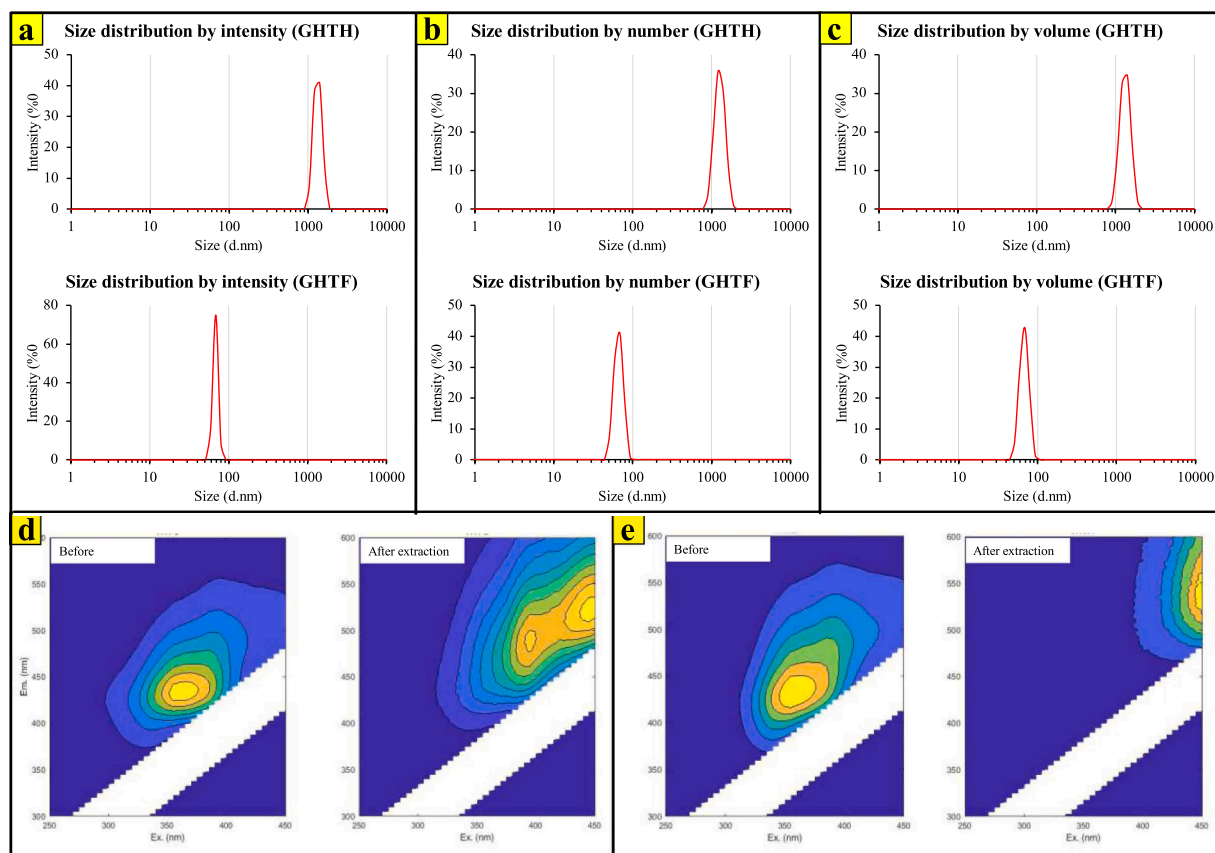


Fig. 7. Size distribution express as function of: (a) density, (b) number, (c) volume for the GHTH and GHTF samples. Fluorescence spectra of GHTF (d), and GHTH (e) as recorded before and after humic acid extraction.

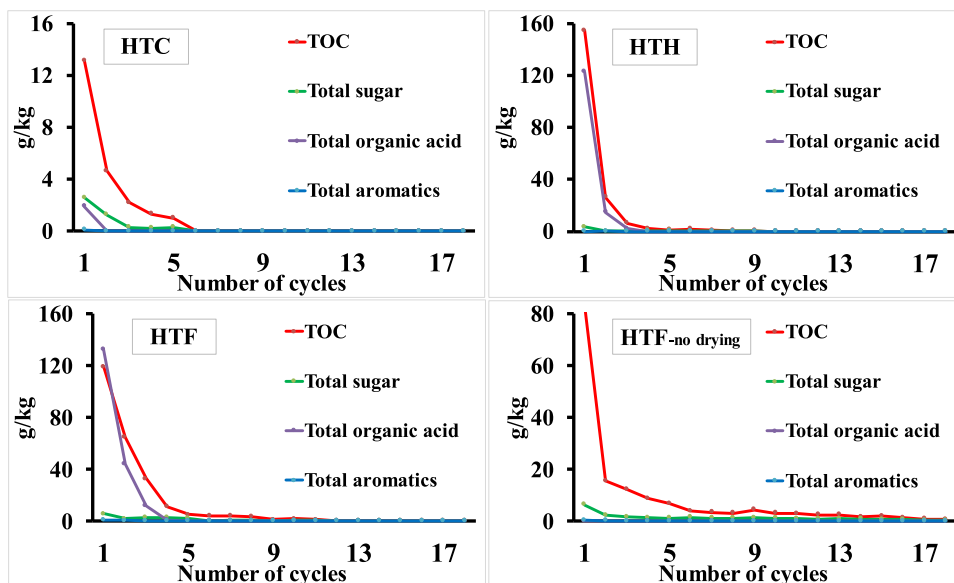


Fig. 8. Extracted leachate from washing process of different hydrochars for total organic carbon, total sugars, organic acids and aromatics (g/kg).

sequential washing confirms that HTC produces the most stable hydrochars with low solubility, while HTH and HTF yield more labile organic fractions capable of enhancing soil organic matter.

Detailed analysis of the dissolved organic constituents, including sugars, organic acids, and aromatics, further elucidated the composition of the leached carbon. GHTC released only trace quantities of these compounds, dominated by small amounts of sugars in the first wash, reinforcing its chemical stability. Conversely, GHTH leachate contained significant organic acid concentrations (125 g/kg) with minor contributions from sugars and aromatics, consistent with the accumulation of acidic humification intermediates. GHTF showed the highest organic acid release (140 g/kg) and detectable aromatics, indicating the presence of small oxygen-rich fulvic-like molecules that dissolve readily in water. Under the HTF-no-drying condition, however, sugars dominated the leachate, while organic acids and aromatics were lower, implying that non-dried hydrochars retained more labile saccharides and fewer recalcitrant oxidation products.

The pH evolution of leachates, across the first five cycles (Fig. 9), further reflected these chemical differences. GHTC exhibited mildly acidic pH values (5.97–6.27), typical of weak carboxylic functional groups. GHTH displayed near-neutral pH values (6.82–6.92), indicating a balance between acidic and basic functionalities. In contrast, GHTF produced markedly alkaline leachates, peaking at pH 9.26 in the third cycle and averaging above 8.5, due to the presence of basic soluble species and residual alkali ions from the fulvification process. The HTF-no-drying condition showed the highest initial pH (9.69), which decreased to 7.68 by cycle five, suggesting that drying favors the formation of weakly acidic groups, while wet storage preserves more reactive, alkaline soluble phases (Esfahani et al., 2015; Harvey et al., 2012; Zhu et al., 2015). Together, the leaching and pH results confirm that HTC hydrochar is the most stable and least soluble, while HTH and HTF generate more reactive, humified, and alkaline fractions. The progressive increase in alkalinity enhances the formation of water-soluble carbon compounds beneficial for soil applications but also necessitates washing when hydrochars are intended for adsorption to ensure chemical stability and prevent contaminant release.

3.5. Adsorption study of CV removal

3.5.1. Screening of hydrochars from different processes for the removal of CV

In preliminary screening experiments, the removal efficiency of crystal violet (CV, 100 mg/L) was assessed using hydrochars derived from the three different hydrothermal processes (GHTC, GHTH, and GHTF) at a dosage of 1 g/L. As shown in Fig. 10, all hydrochars exhibited considerable adsorption capacity, with performance increasing notably from GHTC to GHTF. Specifically, while the GHTC sample achieved an adsorption capacity of 63.05 mg/g, GHTH showed a markedly higher CV removal of 82.03 mg/g. The low standard deviation observed for GHTH ($\pm 0.38\%$) further indicates consistent performance in crystal violet removal, suggesting reliable adsorptive behavior under these conditions. The GHTF hydrochar demonstrated the best performance, achieving adsorption capacity of 87.41 mg/g. Although GHTF exhibited a slightly higher adsorption capacity for crystal violet than GHTH, leachate analysis showed that it released significantly more dissolved components and required multiple washing cycles to ensure safe use. This indicates that GHTH is a more practical adsorbent for dye removal applications, as it offers high adsorption efficiency combined with greater chemical stability in aqueous environments. Therefore, GHTH was selected for the subsequent experiments aimed at a more detailed understanding of the adsorption process and its behavior as an adsorbent.

3.5.2. Adsorption isotherm

Understanding the adsorption mechanism and the interactions between adsorbent and adsorbate relies significantly on the analysis of adsorption isotherms. To investigate the equilibrium adsorption isotherm, experiments were conducted at room temperature using 0.05 g of GHTH in 50 mL of CV solutions, with initial concentrations varying across 10, 20, 50, 100, 200, 350 and 500 mg/L. Taking into account a removal efficiency of 67.68 % and an adsorption capacity of 135.4 mg/g, an initial concentration of 200 mg/L was selected for subsequent experiments (Fig. 11-a). The interaction between the solid surface and the CV molecules, in terms of both surface characteristics and adsorption strength, can be evaluated through the equilibrium adsorption isotherm, which links the concentration of CV in the liquid phase to the amount retained on the solid (Han et al., 2012). Fig. 11-b and Table 4 display the

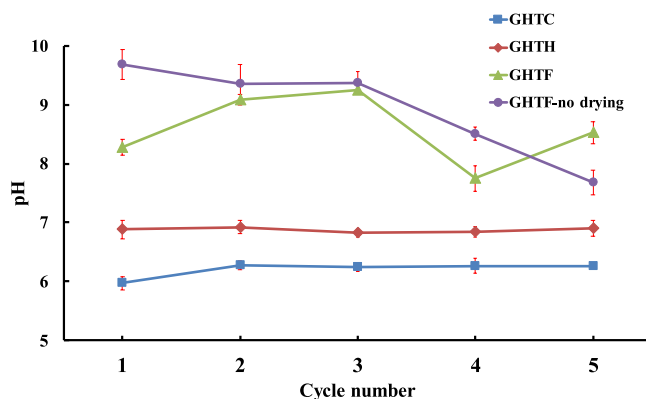


Fig. 9. pH value of leachate obtained from 5 washing cycle of hydrochars from HTC, HTH and HTF processes.

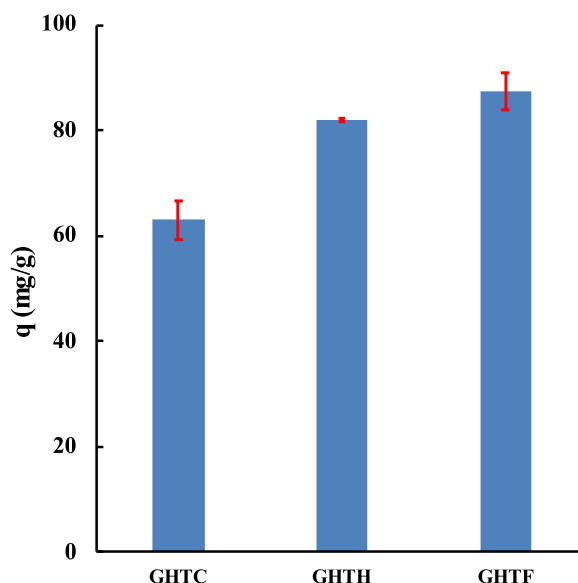


Fig. 10. Adsorption capacity of GHTC, GHTH and GHTF for crystal violet removal (mg/g).

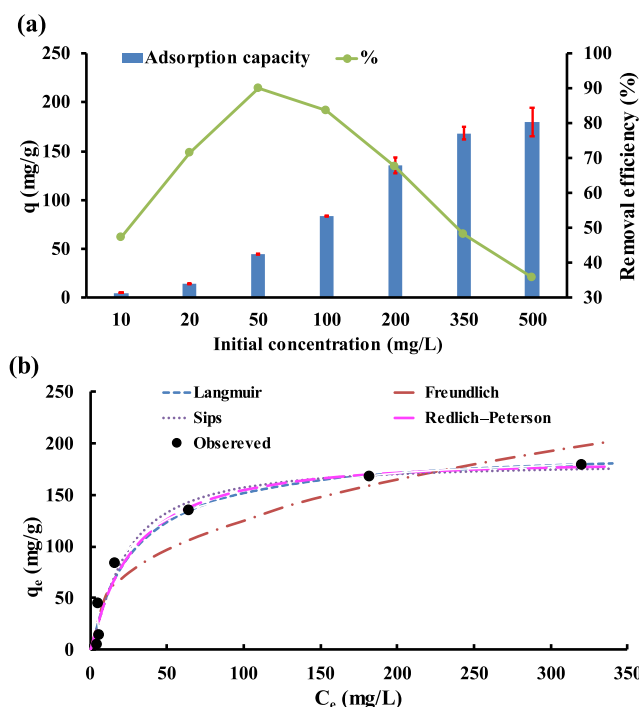


Fig. 11. a) Effect of initial concentration on adsorption capacity of CV removal, b) Langmuir, Freundlich, Sips and Redlich-Peterson isotherm models of adsorption of CV onto GHTH.

parameters obtained from the Langmuir, Freundlich, Sips and Redlich-Peterson isotherm models.

The equilibrium adsorption behavior of CV onto GHTH is well described by monolayer-type isotherms, as indicated by the high correlation coefficients obtained for the Langmuir ($R^2 = 0.9601$, RMSE = 13.32 mg/g) and Sips models ($R^2 = 0.9643$, RMSE = 12.60 mg/g), with the Sips model showing a slightly improved fit. The Langmuir maximum adsorption capacity ($q_{\max} = 196.28$ mg/g) is close to that derived from the Sips model ($q_e = 181.15$ mg/g) and to the experimentally observed values, confirming internal consistency of the equilibrium analysis. These results indicate that the Langmuir model describes the CV adsorption behavior more accurately than the Freundlich model (Kyı et al., 2020), which occurs predominantly via monolayer coverage on energetically similar

Table 4
Isotherm model's parameters.

Model	Parameters	values	R ² (nonlinear)	RMSE (mg/g)
Langmuir	q _{max} (mg/g)	196.28	0.9601	13.32
	K _L (L/g)	0.0343		
Freundlich	K _F (mg/g)	21.75	0.8845	22.65
	n	2.62		
Sips	K _s (L/mg)	0.0211	0.9643	12.60
	β _s	1.25		
	q _s (mg/g)	181.15		
Redlich–Peterson	K _P (L/g)	6.2594	0.9608	13.19
	α (L/mg)	0.0243		
	β	1.05		

surface sites, while allowing for a limited degree of surface heterogeneity (Duraisamy et al., 2015). The strong Langmuir-type adsorption behavior observed for GHTH can be directly linked to its physicochemical characteristics. Although GHTH exhibits a moderate specific surface area compared to GHTF, its surface is enriched with oxygen-containing functional groups, as evidenced by FTIR analysis, which promotes strong interactions with the cationic crystal violet molecules. The presence of mesopores facilitates dye diffusion and accessibility of active sites, while the negatively charged surface under alkaline conditions enhances electrostatic attraction.

3.5.3. Adsorption kinetics

The adsorption kinetics experiments, performed with an initial concentration of 200 mg/L and contact times ranging from 20 min to 24 h, are illustrated in Fig. 12. During the initial 30 min of the adsorption process, a rapid increase in CV sorption was observed, followed by a slower rate of adsorption as the system neared equilibrium. The initial rapid removal of CV is likely due to its diffusion into the mesopores on the surface of GHTH, which has an average pore diameter of 4.1 nm. As the process continued, CV molecules penetrated deeper into the intraparticle matrix, leading to a gradual decrease in the removal rate, primarily caused by the lower dye concentration and the reduced number of available adsorption sites (Toor and Jin, 2012).

To investigate the kinetic mechanism of the adsorption process, the pseudo-first-order, pseudo-second-order models and Weber–Morris intraparticle diffusion were applied, and the findings are summarized in Table 5. The pseudo-second-order model showed a higher coefficient R² value (0.999) compared to the pseudo-first-order model (0.9563), indicating a better fit with the experimental data for CV adsorption onto GHTH. This suggests that the adsorption process is mainly driven by chemisorption, involving electron sharing between GHTH and CV (Aysan et al., 2016). The Weber–Morris model exhibits a poor fit (R² = 0.404) and a substantial intercept (C = 125.91 mg/g), indicating that intraparticle diffusion is not the sole rate-limiting mechanism and that surface adsorption plays a dominant role during CV uptake (Dharmarathna and Priyantha, 2024). The non-zero intercept further suggests the presence of boundary layer effects, reflecting rapid initial surface interaction between CV molecules and active sites on GHTH.

3.5.4. Effect of adsorbent dosage and pH

Five different dosages of GHTH (0.1–10 g) were added to a CV solution with an initial concentration of 200 mg/L to assess the removal efficiency and adsorption capacity, as illustrated in Fig. S12. As the adsorbent dosage increased from 0.1 to 10 g/L, the dye removal efficiency for CV rose from 20 % to 94.5 % for CV. This improvement is likely due to the greater surface area and the corresponding increase in available active sites resulting from the higher adsorbent mass (Lafi et al., 2014). However, increasing the dosage from 1 g/L to 2 g/L, led to only a slight enhancement in removal efficiency, from 67.7 % to 69.5 %. Conversely, the adsorption capacity (q) of GHTH steadily declined from 412.6 mg/g to 18.5 mg/g as the adsorbent dosage increased from 0.1 to 10 g/L. The reduction in the amount of dye adsorbed per unit mass of adsorbent with increasing adsorbent dosage is attributed to a decrease in the concentration gradient or flux between the dye in solution and the adsorbent surface. As the adsorbent mass increases, this gradient diminishes, leading to less dye being adsorbed per gram of adsorbent and, consequently, a lower q value (Vadivelan and Kumar, 2005).

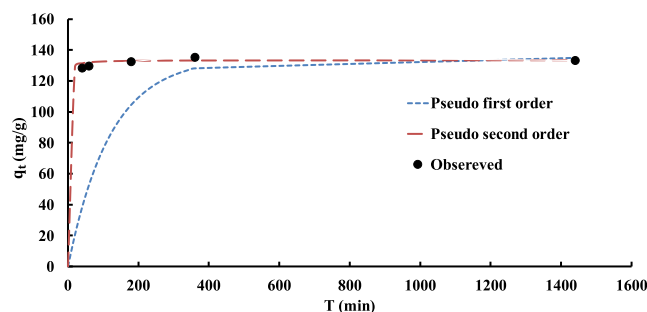


Fig. 12. Pseudo first-order and pseudo second-order kinetic models of CV removal.

Table 5

Kinetic fitting parameters for removal of CV by GHTH adsorbent.

Pseudo-first order model			Pseudo-second order model			Weber-Morris model		
q_e (mg/g)	K_1 (1/min)	R^2	q_e (mg/g)	K_2 (g/mg.min)	R^2	K_{id} (g/mg.min ^{1/2})	C (mg/g)	R^2
13.302	0.0083	0.9563	133.33	0.0117	0.9999	0.270	125.91	0.404

Considering the balance between removal efficiency and adsorption capacity, a dosage of 1 g/L was chosen for the following experimental procedures.

The effect of solution pH on adsorption efficiency was examined (Fig. S13), as pH changes can influence both the ionization of the dye molecules and the surface characteristics of adsorbents (Nandi et al., 2009). Therefore, the adsorption of CV onto GHTH was studied at room temperature using an initial dye concentration of 200 mg/L and a GHTH dose of 1 g/L, with the solution pH adjusted in the range of 3–11. As shown in Fig. 13, the adsorption capacity of CV on GHTH rose from 112.1 to 142.12 mg/g as the pH increased from 3 to 9. Similarly, the percentage removal improved from 56 % at pH 2.0–60.1 % at pH 9.0. A more significant increase in both adsorption capacity and removal efficiency was observed between pH 9 and 11, primarily attributed to changes in the surface charge of GHTH. The highest amount of adsorption capacity and removal efficiency have been observed at pH of 11 with the values of 178.15 mg/g and 89.1 %, respectively. With rising pH levels, the GHTH surface becomes more negatively charged, which enhances the adsorption of the positively charged cationic dye CV through electrostatic attraction (Salleh et al., 2011). The maximum adsorption capacity of 196.28 mg/g from Langmuir model achieved for GHTH is comparable to the values reported for similar adsorbents in previous studies (Table 6).

In summary, compared to GHTF, GHTH was identified as the most suitable adsorbent for crystal violet removal, as it exhibited greater chemical stability and lower leachability while maintaining high adsorption capacity, a strong Langmuir isotherm fit, and chemisorption dominated kinetics. Its performance further improved under alkaline conditions at pH 11, achieving a maximum removal efficiency of 89.1 %.

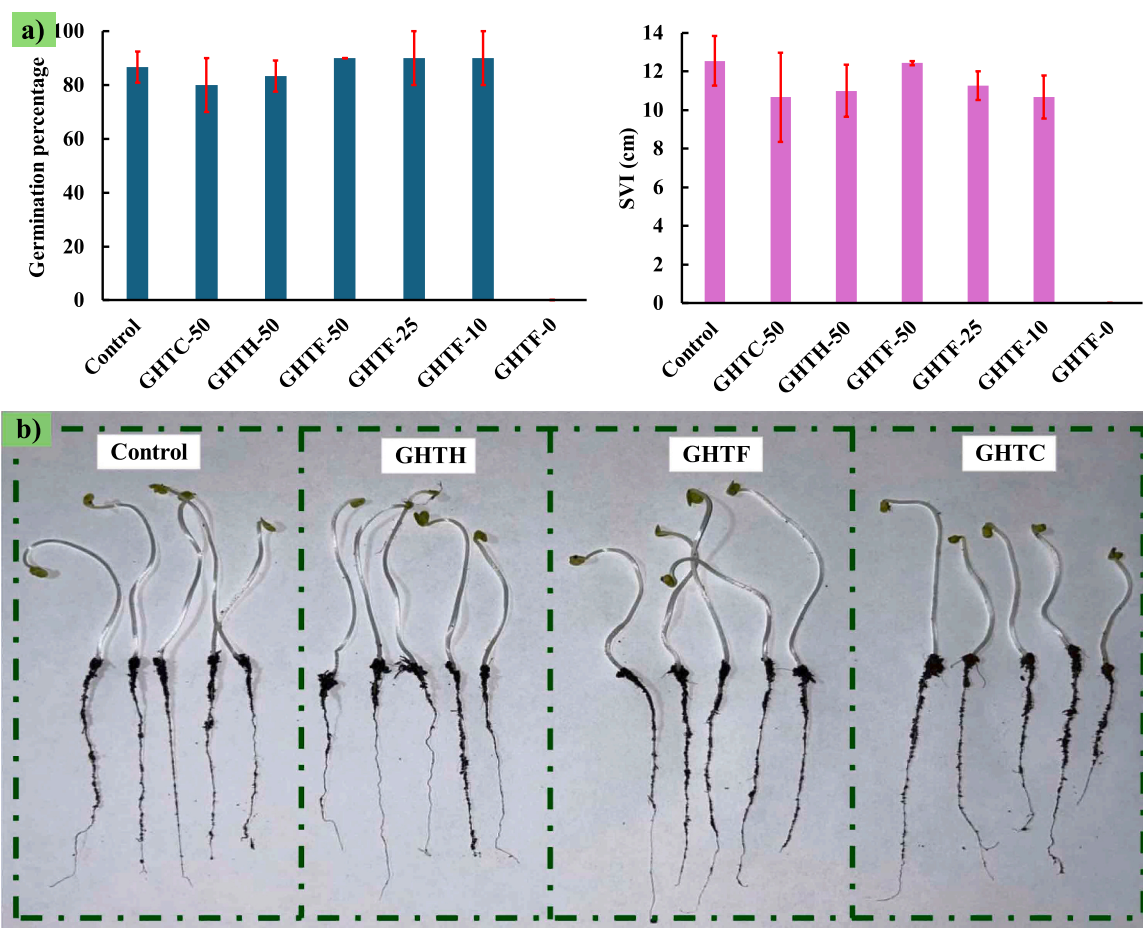


Fig. 13. a) Germination percentage and SVI (cm) of process liquid from HTC, HTH and HTF with different dilution rate from 0 to 50 times, b) Seed roots and shoots of control, GHTC, GHTH and GHTF samples.

Table 6

Comparison of maximum crystal violet adsorption capacities reported for various adsorbents.

Feedstock	Adsorbent type	Adsorption capacity (mg/g)	Temperature (°C)	References
Pomelo peel	Activated biochar	805.69	30	(Tang et al., 2024)
Bamboo leaf	Activated carbon	229.35	32	(Ghosh et al., 2017)
Grass	Hydrochar	196.28	25	This study
Pineapple leaf powder	Biomass	78.22	20	(Chakraborty et al., 2012)
Jackfruit leaf powder	Biomass	43.40	20	(Saha et al., 2012)
Palm kernel shell	Biochar	24.45	25	(Kyi et al., 2020)

3.6. Germination results

Fig. 13-a shows the germination percentage and Seedling Vigor Index (SVI) of process liquids GHTC, GHTH and GHTF under different dilutions. Undiluted GHTF completely inhibited seed germination (0 %), indicating severe phytotoxicity at high concentrations. In contrast, 50-time dilution treatments, including GHTF-50, GHTH-50, and GHTC-50, maintained high germination percentages (80–95 %), demonstrating that dilution substantially reduces inhibitory effects.

The observed response reflects a concentration-dependent balance between inhibitory low-molecular-weight constituents and beneficial humic/fulvic-like dissolved organic matter. The HTF liquid contained high total organic carbon (TOC, 58.53 g/L) and elevated concentrations of organic acids (acetic acid) as well as phenolic derivatives, which are known to suppress seed germination at high doses (Bargmann et al., 2013); this is consistent with the complete inhibition observed for undiluted HTF. Upon dilution (25–50 ×), the inhibitory load is strongly reduced, while humic/fulvic-like fractions remain present, as indicated by pronounced humic-region EEM fluorescence, smaller colloidal particle size, and a more negative zeta potential (−8.72 mV), all suggesting stable and mobile humic/fulvic-like assemblies. In line with this, germination and SVI improved in the sequence HTF > HTH > HTC at the same dilution, supporting that increasing alkalinity redirects the liquid composition toward bioactive humified/fulvified dissolved organic matter rather than simple soluble sugars, which were absent in the HTF liquid phase.

The maximum germination rate of 95 % at 50 × dilution is comparable to or slightly higher than those reported for established commercial biostimulants, including lignin-derived formulations such as KIEMR, which enhanced cucumber seed germination up to 98 ± 0.6 % (Campobenedetto et al., 2020). Comparable germination responses have also been reported for humic-fulvic and amino-acid-based stimulants, with final germination percentages typically ranging between 88 % and 93 % depending on cultivar and formulation (Georgiu et al., 2025), as well as for plant- and microbial-based biostimulants such as Kelpak® and PGPR, which promote germination to values close to 90 % in responsive genotypes (Makhaye et al., 2021). SVI values followed a similar trend, with undiluted GHTF producing the lowest performance and higher dilutions restoring seedling vigor to levels close to those of the control. At a 50-time dilution, SVI was ranked as control > GHTF(50) > GHTH(50) > GHTC(50). These findings are consistent with previous reports where hydrothermal process waters reduced germination and vigor at full strength but recovered upon dilution due to the decreased concentration of phytotoxic compounds such as organic acids and phenolics (Farru et al., 2022).

Fig. 13-b further illustrates these trends by showing seedling photographs, where control and diluted treatments (especially GHTF-50 and GHTH-50) supported normal root and shoot growth, while GHTC-50 suppressed the development. The inhibition of germination observed in undiluted GHTF suggests over-fertilization resulting from the excess of soluble organic carbon, organic compounds, and ions. Upon dilution, these harmful effects diminished, allowing the beneficial effects of compounds produced by the addition of KOH, such as neutralized pH, increased potassium availability, and oxygenated organics, to become more apparent in GHTF and GHTH compared to GHTC.

While the germination assays presented in this study provide a first indication of the bioactivity and phytotoxicity thresholds of hydrothermal process liquids, a more comprehensive assessment of plant-soil interactions is required to fully evaluate their agronomic potential. Future studies should therefore complement germination tests with additional physiological and soil-related analyses, including chlorophyll content, biomass accumulation, and root-shoot development, as well as soil parameters such as moisture retention, electrical conductivity (EC), and cation exchange capacity (CEC). Such measurements would allow a clearer distinction between short-term stimulation effects and longer-term impacts on nutrient availability, salinity stress, and soil chemical buffering (Koprivica et al., 2025). Moreover, extending the experiments to pot and soil incubation trials would enable direct comparison of HTC-, HTH-, and HTF-derived solid and liquid products under realistic growth conditions. Integrating these parameters, as demonstrated in recent studies on humic-based biostimulants, would significantly strengthen the mechanistic understanding and practical relevance of hydrothermal products for sustainable soil and crop management.

3.7. Proposed application for solid and liquid by-product

The solids and liquids derived from HTC, HTH, and HTF displayed distinct physicochemical and functional properties, enabling different valorization strategies. Hydrochar from HTC exhibited the highest heating value (HHV, 23.6 MJ/kg), a direct consequence of dehydration and carbon enrichment during processing, making it a promising candidate for fuel applications where energy density is critical. The thermal stability profile (ash-free $T_{50} = 434.7$ °C) further supports its suitability as a potential solid biofuel, since higher degradation temperatures correlate with improved combustion performance. Hydrochar from HTH showed lower HHV (18.9 MJ/kg) and moderate stability ($T_{50} = 332.4$ °C), but contained abundant oxygenated functional groups and a higher O/C ratio (0.42) relative

to HTC. These features increase polarity and enhance sorption interactions, while the porous structure promotes accessibility, making it suitable as an adsorbent for environmental remediation.

Hydrochar from HTF exhibited a distinct profile. Eighteen-cycles washing released substantially more dissolved organic matter than in other hydrochars, indicating a higher content of soluble residues in the solid phase. Thermogravimetric analysis showed a higher residue content (42.4 %) with moderate ash-free stability ($T_{50} = 388.9\text{ }^{\circ}\text{C}$). The elevated ash content can be regarded as a positive attribute for soil amendment applications, as it reflects increased mineral content and alkalinity that may contribute to nutrient supply and soil pH buffering. This elevated mineral fraction can be attributed to the greater KOH consumption during fulvicification, promoting incorporation of inorganic phases. Combined with washing to reduce phytotoxicity, the mineral-rich composition supports HTF's potential as a soil amendment. Recent studies confirm that humified carbonaceous solids enriched with minerals can enhance cation exchange capacity, water retention, and microbial activity of soil (Ampong et al., 2022).

The relatively low solid yield observed under HTF conditions (13.2 wt.%) arises from the intentional promotion of carbon transfer into the liquid phase rather than solid retention. As a result, this outcome represents a purposeful redistribution of carbon toward high-value humic and fulvic fractions instead of reduced process efficiency. From a carbon distribution perspective, HTF differs fundamentally from conventional HTC. Based on our previous studies, increasing alkalinity progressively suppresses gas-phase formation during hydrothermal processing, reducing carbon losses as gaseous products from approximately 2 % of total carbon in HTC to zero levels under HTF conditions (Marzban et al., 2024). This trend is consistent with the observed increase in total carbon TC and TOC concentrations in the liquid phase from HTC through HTH to HTF, indicating that higher alkalinity preferentially redirects carbon toward liquid-phase products. Consequently, increasing alkalinity enhances the value of liquid by-products, in contrast to HTC where hydrochar is the dominant product and the liquid fraction is typically regarded as a secondary by-product or waste stream. The suppression of gas formation implies that whole of the biomass carbon is retained within the system, predominantly in the liquid phase. The resulting HTF liquid therefore constitutes a value-added product enriched in humic- and fulvic-like carbon (Marzban et al., 2024). This redistribution toward small, mobile, and bioactive carbon forms increases the functional carbon efficiency of the process and supports the economic competitiveness of HTF despite its lower solid yield, as both solid and liquid outputs are suitable for agricultural and soil applications, an outcome not previously reported for conventional HTC liquids.

The process liquids also showed valorization potential. Although direct use may be limited by organic acids and aromatics, dilution reduced phytotoxicity. Germination improved at 10 % dilution, consistent with humic- and fulvic-like substances detected by fluorescence spectroscopy (E_4/E_6 ratios of 12–15). Zeta potential values (+4.5 to +5.6 mV) support their role as biostimulants when applied at controlled concentrations. This aligns with studies showing that artificial humic substances improve nutrient uptake, root development, and plant growth under field conditions (Yuan et al., 2022), and serve broader biostimulant functions in improving stress resilience and fertilizer efficiency (Canellas et al., 2022). From a regulatory perspective, the presence of aromatic compounds in hydrothermal products warrants careful consideration depending on the intended application and physical state of the material. For solid carbonaceous materials, quality frameworks such as the European Biochar Certificate (EBC) define strict limits for potentially hazardous aromatics (e.g., $\Sigma 16$ EPA PAHs < 12 mg/kg for basic grade) (EBC, 2021), illustrating how environmental safety is typically managed for biochar-like products. Furthermore, the accumulation of inorganic nutrients and trace metals in hydrochars, such as K, Ca, Cu, Fe, Pb, and Cr, should be carefully evaluated prior to soil application, as hydrothermal processing can promote solid-phase enrichment of these elements, which may be beneficial or restrictive depending on compliance with fertilizer and soil-amendment regulations (Moloeznik Paniagua et al., 2024). In contrast, no universal threshold exists for total dissolved organic matter in liquid biostimulants, as regulatory guidelines generally address specific compounds rather than bulk organic content. For example, the United States Environmental Protection Agency (USEPA) regulates phenolic compounds in water-use contexts at low concentrations, with guideline values for phenol in drinking water on the order of 0.3 mg/L (Mhlongo et al., 2024), highlighting the importance of dilution and controlled application. Accordingly, the diluted use of HTF liquids adopted in this study represents a practical risk-mitigation strategy, while comprehensive compound-specific toxicity and regulatory assessments would be required prior to large-scale or unrestricted environmental deployment.

Although the HTF process requires a relatively high alkaline input (66 wt.% on a dry solid basis), several strategies can improve its economic feasibility at larger scales. Previous studies on alkaline hydrothermal biomass processing have demonstrated that alkali consumption can be significantly reduced through the recovery and reuse of base-rich process liquids or by integrating separation-based recycling approaches, thereby lowering operational costs (Pismenskaya et al., 2022). In this context, the KOH used in the present study should be replaced or partially substituted with cheaper and more economically feasible alternatives. Low-cost alkaline sources, such as biomass- or plant-derived ash, have been reported as viable substitutes for pure alkali reagents in thermochemical and hydrothermal systems, providing sufficient alkalinity while simultaneously contributing additional mineral value (Bedov et al., 2025). Alkaline materials such as agricultural lime have also been investigated; although they typically result in slower formation rates of artificial humic acids (Ghaslani et al., 2024), their use in combination with KOH represents a promising strategy to reduce overall alkali consumption. Furthermore, membrane-based separation concepts, such as electrodialysis or bipolar membrane electrodialysis, have been applied in related bioprocessing contexts to enable selective recovery of alkaline streams alongside organic products, highlighting the potential for downstream alkali recycling without chemical neutralization (Molnár et al., 2010). The application of heterogeneous alkaline or industrial catalysts that can be recovered and reused provides an additional pathway for improving process economics. Considering that HTF targets high-value liquid products such as humic- and fulvic-like biostimulants rather than bulk materials, the implementation of alkali recovery schemes or alternative alkaline sources is regarded as both technically and economically feasible.

4. Conclusions

This work provides a comprehensive and unified assessment of how alkalinity fundamentally reshapes hydrothermal conversion pathways of grass biomass and, in doing so, determines both carbon distribution and product functionality. By systematically comparing hydrothermal carbonization, humification, and fulvification under identical reaction conditions, the study reveals that alkalinity is not merely a process modifier but a decisive control parameter that governs whether biomass carbon is retained as a stable solid, transformed into adsorption-active humified materials, or mobilized into bioactive and colloiddally stable liquid fractions. This mechanistic clarity represents a key advancement beyond conventional HTC-focused studies, which typically optimize a single product without considering cross-phase interactions or functional trade-offs.

Detailed physicochemical analyses demonstrate that HTC favors the formation of energy-dense and intrinsically stable hydrochar dominated by dehydration and condensation reactions, whereas increasing alkalinity progressively promotes depolymerization, oxygenation, and carbon transfer into the liquid phase. Under moderate alkalinity, hydrothermal humification yields a hydrochar with optimized surface chemistry and minimal leachability, translating into superior adsorption performance toward a model cationic dye. Under strong alkalinity, hydrothermal fulvification redirects most of the carbon into the aqueous phase, generating highly dispersed, negatively charged humic- and fulvic-like assemblies that exhibit clear bioactivity at appropriate dilution levels. Importantly, this study demonstrates for the first time that these distinct functionalities can be accessed within a single, tunable hydrothermal platform, without changing feedstock or operating temperature, simply by controlling alkalinity.

Beyond individual performance metrics, the broader significance of this work lies in its integrated valorization concept. By jointly evaluating solid and liquid products in relation to their end-use performance, the study moves beyond the traditional fuel-oriented paradigm of hydrothermal processing and highlights new opportunities for coupling water remediation and soil-related applications within a circular carbon framework. The demonstrated leaching stability of hydrochars, combined with the bioactivity of diluted process liquids, supports a complementary use of both fractions and reduces the notion of “waste streams” in hydrothermal systems.

At the same time, the study intentionally positions germination assays as an initial screening tool rather than a full agronomic assessment. Future work should therefore expand the biological evaluation through pot and soil-based experiments, incorporating quantitative plant physiological indicators such as chlorophyll content, biomass accumulation, and root architecture, alongside soil properties including water-holding capacity, electrical conductivity, and cation and anion exchange capacities. Long-term studies addressing nutrient dynamics, microbial responses, and field-scale performance, will be essential to fully assess the agronomic relevance and environmental safety of both solid and liquid hydrothermal products. Integrating these analyses will further strengthen the translation of alkalinity-tuned hydrothermal systems toward practical applications in sustainable agriculture, water treatment, and circular bioeconomy strategies.

Author's contributions

Saman Ghobadian: Design of experiments, methodology, visualization, data collection, investigation, and writing the original draft of the manuscript. **Oswaldo Romero Romero:** Supervision, methodology, review, and editing. **Matthias Kraume:** Supervision, review, and editing. **Giulia Ischia:** Interpretation, review, and editing. **Thomas Hoffmann:** Interpretation, review, and editing. **Giacomo Rossi:** Interpretation, review, and editing. **Nader Marzban:** Supervision, conceptualization, design of experiment, methodology, writing the original draft of the manuscript - review and editing. All authors have read and approved the final version of the manuscript.

CRediT authorship contribution statement

Giacomo Rossi: Writing – review & editing, Visualization, Software. **Matthias Kraume:** Writing – review & editing, Supervision. **Thomas Hoffmann:** Writing – review & editing, Resources. **Nader Marzban:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Saman Ghobadian:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Romero Oswaldo:** Writing – review & editing, Supervision, Methodology. **Giulia Ischia:** Writing – review & editing, Formal analysis.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.eti.2026.104836](https://doi.org/10.1016/j.eti.2026.104836).

Data availability

No data was used for the research described in the article.

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